Assignment 0

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CS 584 - S26

PART A

$$p = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \quad q = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}, \quad r = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

1. $3p + 2q \Rightarrow \quad 3p = \begin{bmatrix} 6 \\ -3 \\ 12 \end{bmatrix} + 2q = \begin{bmatrix} 0 \\ 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 22 \end{bmatrix}$

2. $\|p\| = \sqrt{2^2 + (-1)^2 + (4)^2} = \sqrt{4 + 1 + 16} = \sqrt{21}$

$$\hat{p} = \frac{1}{\sqrt{21}} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

3. $\|p\| = \sqrt{21}$

The angle θ with y axis is

$$\cos\theta = \frac{p \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}{\|p\|} = \frac{\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}}{\sqrt{21}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{-1}{\sqrt{21}}\right)$$

4. $\cos\alpha = \frac{2}{\sqrt{21}}, \quad \cos\beta = \frac{-1}{\sqrt{21}}, \quad \cos\gamma = \frac{4}{\sqrt{21}}$

5. $p \cdot q = 2(0) + (-1)(3) + (4)(5) = 17, \quad \|q\| = \sqrt{0^2 + 3^2 + 5^2} = \sqrt{34}$

$$\cos\theta = \frac{17}{\sqrt{21} \cdot \sqrt{34}} = \sqrt{\frac{17}{42}}, \quad \theta = \cos^{-1}\sqrt{\frac{17}{42}}$$

6. $p \cdot q = q \cdot p = 17$

7. $p \cdot q = \|p\| \|q\| \cos\theta = \sqrt{21} \cdot \sqrt{34} \cdot \dfrac{17}{\sqrt{21} \cdot \sqrt{34}} = 17$

8. Scalar projection $= q \cdot \hat{p} = \dfrac{q \cdot p}{\|p\|} = \dfrac{17}{\sqrt{21}}$

9. Take any vector v, s.t. $p \cdot v = 0$

$$v = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

10. $$p \times q = \begin{bmatrix} (-1)(5) - 4(3) \\ -(2)(5) - 4(0) \\ (2)(3) - (-1)(0) \end{bmatrix} = \begin{bmatrix} -17 \\ -10 \\ 6 \end{bmatrix}$$

$$q \times p = -(p \times q) = \begin{bmatrix} 17 \\ 10 \\ -6 \end{bmatrix}$$

11. $$p \times q = \begin{bmatrix} -17 \\ -10 \\ 6 \end{bmatrix}$$

12. solving $ap + bq + cr = 0$

- System yields trivial solution $a = b = c = 0$

$\Rightarrow$ p, q, r are linearly independent

13. $p^T q = 17$ (scalar dot product)

$$pq^T = \begin{bmatrix} 2.0 & 2.3 & 2.5 \\ -1.0 & -1.3 & -1.5 \\ 4.0 & 4.3 & 4.5 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 10 \\ 0 & -3 & -5 \\ 0 & 12 & 20 \end{bmatrix}$$

PART B

$$X = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 3 & 2 & -2 \end{bmatrix}, \quad Y = \begin{bmatrix} 4 & -1 & 2 \\ 3 & 0 & -3 \\ 1 & 2 & 1 \end{bmatrix}, \quad Z = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 4 & 5 \\ 3 & 1 & 2 \end{bmatrix}, \quad S = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$$

1. $$X + 2Y = \begin{bmatrix} 10 & -1 & 4 \\ 5 & 3 & -2 \\ 5 & 6 & 0 \end{bmatrix}$$

2. $$XY = \begin{bmatrix} 11 & -2 & 1 \\ 9 & 9 & -7 \\ 16 & -7 & -2 \end{bmatrix}$$

$$YX = \begin{bmatrix} 15 & 5 & -8 \\ -3 & -3 & 6 \\ 3 & 9 & 6 \end{bmatrix}$$

3. $$(XY)^T = \begin{bmatrix} 11 & 9 & 16 \\ -2 & 9 & -7 \\ 1 & -7 & -2 \end{bmatrix}$$

Finding $\rightarrow$ $$Y^T = \begin{bmatrix} 4 & 3 & 1 \\ -1 & 0 & 2 \\ 2 & 3 & 1 \end{bmatrix}, \quad X^T = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 3 & 2 \\ 0 & 4 & -2 \end{bmatrix}$$

$$Y^T X^T = \begin{bmatrix} 11 & 9 & 16 \\ -2 & 9 & -7 \\ 1 & -7 & -2 \end{bmatrix}$$

4. $|X|$ and $|Z|$

Along First row : $|X| = 2 \cdot \det\begin{bmatrix} 3 & 4 \\ 2 & -2 \end{bmatrix} - 1 \cdot \begin{bmatrix} -1 & 4 \\ 3 & -2 \end{bmatrix} + 0 \cdot \det\begin{bmatrix} -1 & 3 \\ 3 & 2 \end{bmatrix}$

- $\det\begin{bmatrix} 3 & 4 \\ 2 & -2 \end{bmatrix} = (3)(-2) - 4(2) = -6 - 8 = -14$
- $\det\begin{bmatrix} -1 & 4 \\ 3 & -2 \end{bmatrix} = (-1)(-2) - (4)(3) = -10$

$\therefore \; |X| = 2(-14) - 1(-10) + 0 = -18$

$|Z|$:

Expand along first row $\rightarrow$

- $2 \cdot \det\begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix} - 0 \cdot \det\begin{bmatrix} 1 & 5 \\ 3 & 2 \end{bmatrix} + (-1) \cdot \det\begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$

- $\det\begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix} = (4)(2) - (5)(1) = 8 - 5 = 3$
- $\det\begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} = (1)(1) - (4)(3) = 1 - 12 = -11$

$|Z| = 2(3) - 0 + (-1)(-11) = 6 + 11 = 17$

5. • For X :

Row 1 · Row 2 : $(2)(-1) + (1)(3) + (0)(4) = -2 + 3 + 0 = 1 \neq 0$

Not orthogonal

- For Y:

Row 1 . Row 2 : $(4)(3) + (-1)(0) + (2)(-3) = 12 + 0 - 6 = 6 \neq 0$

Not orthogonal

- For Z:

Row 1 . Row 2 : $(2)(1) + (0)(4) + (-1)(5) = 2 + 0 - 5 = -3 \neq 0$

Not orthogonal

None of them orthogonal

6. Compute X^{-1} :

$|X| = -18 \neq 0$, so invertible

① Matrix of minors :

$$\begin{bmatrix} \det\begin{bmatrix} 3 & 4 \\ 2 & -2 \end{bmatrix} & \det\begin{bmatrix} -1 & 4 \\ 3 & -2 \end{bmatrix} & \det\begin{bmatrix} -1 & 3 \\ 3 & 2 \end{bmatrix} \\ \det\begin{bmatrix} 1 & 0 \\ 2 & -2 \end{bmatrix} & \det\begin{bmatrix} 2 & 0 \\ 3 & -2 \end{bmatrix} & \det\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \\ \det\begin{bmatrix} 1 & 0 \\ 3 & 4 \end{bmatrix} & \det\begin{bmatrix} 2 & 0 \\ -1 & 4 \end{bmatrix} & \det\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} -14 & -10 & -11 \\ -12 & -4 & 1 \\ 4 & 8 & 7 \end{bmatrix}$$

Cofactor matrix :

$$\begin{bmatrix} +(-14) & -(-10) & +(-11) \\ -(-2) & +(-4) & -(1) \\ +(4) & -(8) & +(7) \end{bmatrix} = \begin{bmatrix} -14 & 10 & -11 \\ 2 & -4 & -1 \\ 4 & -8 & 7 \end{bmatrix}$$

Adjugate :

$$\text{adj}(X) = \begin{bmatrix} -14 & 2 & 4 \\ 10 & -4 & -8 \\ -11 & -1 & 7 \end{bmatrix}$$

- $X^{-1} = \frac{1}{|X|} \text{adj}(X) = -\frac{1}{18} \begin{bmatrix} -14 & 2 & 4 \\ 10 & -4 & -8 \\ -11 & -1 & 7 \end{bmatrix} = \begin{bmatrix} 7/9 & -1/9 & -2/9 \\ -5/9 & 2/9 & 4/9 \\ 11/18 & 1/18 & -7/18 \end{bmatrix}$

- $Y^{-1} = \frac{1}{42} \begin{bmatrix} 6 & 5 & 3 \\ -6 & 2 & 18 \\ 6 & -9 & 3 \end{bmatrix} = \begin{bmatrix} 1/7 & 5/42 & 1/14 \\ -1/7 & 1/21 & 3/7 \\ 1/7 & -3/14 & 1/14 \end{bmatrix}$

7. $Z^{-1} = \frac{1}{17} \begin{bmatrix} 3 & -1 & 4 \\ 13 & 7 & -11 \\ -11 & -2 & 8 \end{bmatrix} = \begin{bmatrix} 3/17 & -1/17 & 4/17 \\ 13/17 & 7/17 & -11/17 \\ -11/17 & -2/17 & 8/17 \end{bmatrix}$

8. $Xs = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 13 \\ 5 \end{bmatrix}$

9. $\|s\| = \sqrt{(-1)^2 + 4^2 + 0^2} = \sqrt{17}$, so $\hat{s} = \frac{1}{\sqrt{17}} \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$

Scalar projection of a row vector v is $v \cdot \hat{s}$

- Row 1 of X: $[2, 1, 0] \cdot \hat{s} = \frac{2(-1) + 1(4) + 0}{\sqrt{17}} = \frac{-2+4}{\sqrt{17}} = \frac{2}{\sqrt{17}}$
- Row 2 of X: $[-1, 3, 4] \cdot \hat{s} = \frac{(-1)(-1) + 3(4) + 4(0)}{\sqrt{17}} = \frac{13}{\sqrt{17}}$
- Row 3 of X: $[3, 2, -2] \cdot \hat{s} = \frac{3(-1) + 2(4) + (-2)(0)}{\sqrt{17}} = \frac{5}{\sqrt{17}}$

10. Row 1: $\left(\frac{2}{\sqrt{17}}\right) \cdot \frac{1}{\sqrt{17}} \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} = \frac{2}{17} \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -2/17 \\ 8/17 \\ 0 \end{bmatrix}$

Row 2: $\left(\frac{13}{\sqrt{17}}\right) \cdot \frac{1}{\sqrt{17}} \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -13/17 \\ 52/17 \\ 0 \end{bmatrix}$

Row 3: $\left(\frac{5}{\sqrt{17}}\right) \cdot \frac{1}{\sqrt{17}} \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -5/17 \\ 20/17 \\ 0 \end{bmatrix}$

11. Columns of X:

$c_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, $c_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$, $c_3 = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$

Linear combination: $s_1 c_1 + s_2 c_2 + s_3 c_3 = (-1) c_1 + 4 c_2 + 0 c_3 = \begin{bmatrix} 2 \\ 13 \\ 5 \end{bmatrix}$

12. $t = Y^{-1}s$

$$Y^{-1} = \begin{bmatrix} 1/7 & 5/42 & 1/14 \\ -1/7 & 1/21 & 3/7 \\ 1/7 & -3/14 & 1/14 \end{bmatrix}, \quad s = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$$

Verification: $$Yt = \begin{bmatrix} 4 & -1 & 2 \\ 3 & 0 & -3 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \\ -1 \end{bmatrix} = \begin{bmatrix} 4/3 - 1/3 - 2 \\ 1 + 0 + 3 \\ 1/3 + 2/3 - 1 \end{bmatrix} = \begin{bmatrix} 1-2 \\ 4 \\ 1-1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$$
$$= s$$

$$\therefore \quad t = \begin{bmatrix} 1/3 \\ 1/3 \\ 1 \end{bmatrix}$$

13. $t = Z^{-1}s$

$$Z^{-1} = \begin{bmatrix} 3/17 & -1/17 & 4/17 \\ 13/17 & 7/17 & -11/17 \\ -11/17 & -2/17 & 8/17 \end{bmatrix}, \quad s = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$$

- $t_1 = -7/17$, $t_2 = 15/17$, $t_3 = 3/17$

Reason: $|Z| = 17 \neq 0$, so Z is invertible. Thus, $Zt = s$ has a unique solution $t = Z^{-1}s$

PART C

$$M = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}, \quad N = \begin{bmatrix} 5 & -3 \\ -3 & 6 \end{bmatrix}, \quad P = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix}$$

1. $\det(M - \lambda I) = \begin{vmatrix} 3-\lambda & 2 \\ -1 & 4-\lambda \end{vmatrix} = (3-\lambda)(4-\lambda) - 2(-1)$

$= 12 - 7\lambda + \lambda^2 + 2 = \lambda^2 - 7\lambda + 14$

$$\lambda^2 - 7\lambda + 14 = 0 \Rightarrow \lambda = \frac{7 \pm i\sqrt{7}}{2}$$

Finding eigenvectors

- For $\lambda = \dfrac{7 + i\sqrt{7}}{2}$: $(M - \lambda I)v = 0 \Rightarrow v_2 = \dfrac{1 + i\sqrt{7}}{4} v_1$

Let $v_1 = 4$, then $v_2 = 1 + i\sqrt{7}$:

$$v_1 = \begin{bmatrix} 4 \\ 1 + i\sqrt{7} \end{bmatrix}$$

- For $\lambda = \dfrac{7 - i\sqrt{7}}{2}$:

$$v_2 = \begin{bmatrix} 4 \\ 1 - i\sqrt{7} \end{bmatrix}$$

2. $v_1 \cdot v_2 = (4)(4) + (1 + i\sqrt{7})(1 - i\sqrt{7}) = 16 + (1 + 7) = 16 + 8 = 24$

3. $\det(N - \lambda I) = (5-\lambda)(6-\lambda) - 9 = \lambda^2 - 11\lambda + 21$

$$\lambda = \frac{11 \pm \sqrt{37}}{2}$$

Eigenvectors

- For $\lambda_1 = \frac{11+\sqrt{37}}{2}$:

$$v_1 = \begin{bmatrix} 6 \\ -1-\sqrt{37} \end{bmatrix}$$

- For $\lambda_2 = \frac{11-\sqrt{37}}{2}$:

$$v_2 = \begin{bmatrix} 6 \\ -1+\sqrt{37} \end{bmatrix}$$

$$v_1 \cdot v_2 = 6(6) + (-1-\sqrt{37})(-1+\sqrt{37}) = 36 + (1-37) = 0$$

4. Key Property:

Matrix N is symmetric ($N^T = N$). A fundamental property of real symmetric matrices is that eigenvectors corresponding to distinct eigenvalues are orthogonal.

Reason:

This follows from the Spectral Theorem, which states that every real symmetric matrix is orthogonally diagonalizable. Thus, its eigenvectors form an orthogonal basis.

$\therefore$ Eigenvectors corresponding to distinct eigenvalues are orthogonal because N is symmetric.

5. Trivial Solution

$$t = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

6. Solving $Pt = 0$

$$P = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} \Rightarrow 2t_1 + 4t_2 = 0 \Rightarrow t_1 = -2t_2$$

Non-trivial Solutions

- let $t_2 = 1$: $t_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$
- Let $t_2 = 2$: $t_2 = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$

7. $\det(M) = (3)(4) - (2)(-1) = 12 + 2 = 14 \neq 0 \Rightarrow M$ is invertible

$$t = M^{-1} \cdot 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

If M is invertible, the equation $Mt = 0$ has a unique solution because M^{-1} exists.

PART D

$$f(x) = 2x^2 - 1, \quad g(x) = 3x^2 + 4, \quad h(x,y) = x^2 + y^2 + xy$$

1. $f'(x) = \dfrac{d}{dx}(2x^2 - 1) = 2 \cdot 2x - 0 = 4x$

$f''(x) = 4x$:

$$\frac{d}{dx}(4x) = 4$$

2. $$\frac{\partial h}{\partial x} = \frac{\partial}{\partial x}(x^2+y^2+xy) = 2x + 0 + y = 2x+y$$

$$\frac{\partial h}{\partial y} = \frac{\partial}{\partial y}(x^2+y^2+xy) = 0 + 2y + x = 2y+x$$

3. $$\nabla h(x,y) = \begin{bmatrix} \frac{\partial h}{\partial x} \\ \frac{\partial h}{\partial y} \end{bmatrix}$$

$$= \begin{bmatrix} 2x + y \\ 2y + x \end{bmatrix}$$

4. <u>Without</u> <u>chain rule</u>

Substitute $g(x) = 3x^2+4$ into $f(x)$:

$$f(g(x)) = 2(3x^2+4)^2 - 1$$

Expanding $(3x^2+4)^2$:

$$(3x^2+4)^2 = 9x^4 + 24x^2 + 16$$

Multiply by 2 and subtract 1:

$$f(g(x)) = 2(9x^4 + 24x^2 + 16) - 1$$
$$= 18x^4 + 48x^2 + 31$$

$$\frac{d}{dx} f(g(x)) = \frac{d}{dx}(18x^4 + 48x^2 + 31)$$

$$= 72x^3 + 96x$$

With Chain rule :

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

- $f'(u)$ where $u = g(x)$:

  $f'(u) = 4u \Rightarrow f'(g(x)) = 4(3x^2+4)$

- computing $g'(x)$:

  $g'(x) = 6x$

$$\Rightarrow \frac{d}{dx} f(g(x)) = 4(3x^2+4) \cdot 6x = 24x(3x^2+4) = 72x^3 + 96x$$